

THERMAL IMAGING CAMERA · CBT SERIES

320 × 240 OLEI CBT-3C THERMAL CAMERA

A 320 × 240 thermal imaging camera with a 12 μm detector pitch, 30 Hz frame rate and a configurable high-temperature alarm, on a single USB Type-C cable and a 1/4-inch tripod thread.



■ WHY THIS SENSOR

- **320 × 240 resolution at 12 μm pitch** — 30 Hz image frequency across a 32° × 24° field of view
- **0.1 °C minimum resolution, ±0.5 °C accuracy** — over a 30–45 °C measurement range
- **External ambient temperature sensor** — temperature compensation referenced to the room, not assumed
- **Configurable high-temperature alarm** — with automatic image capture on the alarm event
- **Three palettes** — WhiteHot, Iron, Rainbow — pick the one that makes the gradient you care about legible
- **Single USB Type-C connection** — one cable for power and data to a Windows 7 or later host
- **1/4-inch thread mount** — standard tripod and bracket hardware, 68 × 68 × 60 mm, 350 g
- **Specified for a still, controlled room** — 16–32 °C windless test environment at 1.5–2 m

320 × 240

RESOLUTION

±0.5 °C

ACCURACY

30 Hz

IMAGE FREQUENCY

32° × 24°

FIELD OF VIEW

■ APPLICATION SCENARIOS



THERMAL INSPECTION

Non-contact surface temperature in a fixed frame



PROCESS MONITORING

Hot-spot alarms on equipment and enclosures



INDUSTRIAL AUTOMATION

Guarded cells, robot envelopes and access ways



AREA MONITORING

Perimeter and equipment-envelope intrusion detection

CBT-3C — SPECIFICATIONS

PARAMETER	TECHNICAL DATA
Resolution	320 × 240
Detector pitch	12 μm
Image frequency	30 Hz
Field of view	H 32° × V 24°
Palette	WhiteHot, Iron, Rainbow
Temperature range	30 to 45 °C
Minimum resolution	0.1 °C
Accuracy	±0.5 °C
Test environment	16 to 32 °C, windless
Test distance	1.5 to 2 m
Temperature compensation	External ambient temperature sensor

PARAMETER	TECHNICAL DATA
Alarm	High temperature alarm, adjustable
Photograph	High temperature image capture
Communication	USB Type-C
Installation	1/4-inch thread
Size	68 × 68 × 60 mm
Weight	350 g
Storage temperature	-20 to 50 °C
Humidity	20% to 80% RH
Platform	Windows 7 and above
Detection algorithm	Single-subject region detection

■ CONDITIONS THE ACCURACY DEPENDS ON

WORKING DISTANCE 1.5 to 2 m from the subject or surface — closer or further invalidates the ±0.5 °C figure

AIR MOVEMENT Still air; HVAC outflow and open doorways shift readings

WARM - UP Let the camera reach thermal equilibrium in the room before relying on readings

MEASUREMENT RANGE 30 to 45 °C only — this is not a general industrial pyrometer

AMBIENT TEMPERATURE 16 to 32 °C at the camera and the target

AMBIENT SENSOR The external ambient temperature sensor must be connected — it is part of the specification

FRAME CONTENT Keep strong heat sources and direct sunlight out of frame; one subject region at a time

WHAT THE READING IS A surface temperature under the above conditions — never a body temperature or a health determination

Certification. Confirm laser-free classification, EMC and regional approval status with MorpheusTEK before purchase — the manufacturer's datasheet for this product lists no EMC, safety or regional approval standards.

Not a medical device. This camera is a thermal imaging instrument with a configurable high-temperature alarm. It is **not** FDA-cleared or approved for measuring human body temperature, for fever screening, or for any diagnostic, screening or triage purpose, and it must not be marketed, sold or used for those purposes. In the United States, thermal imaging systems intended for body-temperature measurement are regulated medical devices; the COVID-era enforcement discretion that once covered them has lapsed. Any measurement this camera reports is a surface temperature under the stated test conditions, not a body temperature, and must not be used to make a health determination about any person.

Notes. Every accuracy figure is conditional on the stated test setup — a 16–32 °C windless environment at 1.5–2 m, with the external ambient sensor connected. Outside those conditions, treat readings as indicative only. The 30–45 °C measurement range is narrow: this instrument is not a general industrial pyrometer and will not read hot equipment. The source datasheet is dated 2021; confirm current availability and specification with MorpheusTEK before quoting.

Installation. Mount on the 1/4-inch thread on a stable tripod or bracket at the stated 1.5–2 m working distance, in still air, and connect the external ambient temperature sensor — it is part of the accuracy specification, not an accessory. Let the camera reach thermal equilibrium in the room before relying on readings, keep it out of direct sunlight and away from HVAC outflow, and keep strong heat sources out of frame.

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